Deep Feature Learning From Electromyographic Signals for Gesture Recognition Systems

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***Abstract*— Deep learning applied to electromyography (EMG) signals enables accurate hand gesture recognition, revolutionizing diverse applications such as human-machine interaction, neural interfaces, and rehabilitative robotics. A well-designed deep learning architecture is crucial for accurately and robustly modeling and decoding the multidimensional information embedded in the EMG data. This survey presents a comprehensive review of state-of-the-art deep learning models and, for the first time, offers a categorization of advanced architectures from the perspective of data representations. EMG, as a distinctive biosignal modality, can be characterized through multiple representational forms, including temporal waveforms, spatial images, spectral domains, and graph-based structures comprising interconnected nodes. Consequently, the optimal model architecture is closely tied to the specific data representation employed. In addition, the limited availability of EMG datasets, particularly those with high-quality labels, remains a critical bottleneck and continues to impede the translation of research advances into widespread real-world applications. We therefore examine emerging semi-supervised and self-supervised learning frameworks, which serve as complementary approaches to fully supervised paradigms. Finally, we outline promising future directions for the development of generalizable and robust deep learning for practical EMG decoding.**

***Index Terms*— Electromyography (EMG), deep learning, feature representation, gesture recognition.**

I. Introduction

OVER the past decade, significant advances in electromyography (EMG)-based gesture recognition systems have driven notable progress in applications for human–machine interaction (HMI) [1], [2], [3], [4], [5], assistive technologies [6], [7], [8] and rehabilitative robotics [9], [10], [11]. As presented in Fig. 1, these systems use EMG sensors [12], [13], [14], [15], [16], [17], [18] to capture muscular activity and, coupled with advanced signal analysis, decode gestures. Traditional approaches to EMG analysis employ mainly hand-crafted feature extraction and classical machine learning algorithms [19], [20], [21], [22], but recent breakthroughs in deep learning have revolutionized the field by enabling automatic feature extraction and end-to-end computational learning [21], [23], [24], [25], [26], [27].

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As a branch of representation learning, deep learning aims to extract high-level features directly from the EMG input through a series of computational decoders [27]. EMG signals inherently encapsulate multi-domain information, including spatio-temporal waveforms that capture muscle activation dynamics, frequency-domain features that reveal spectral characteristics of muscle activity, and intrinsic spiking information reflecting motor unit activity. Crucially, the design of deep learning decoders is not considered in isolation; their architecture and performance are intertwined with the quality and nature of EMG representations. Specifically, the choice of representation profoundly influences how the decoders process and interpret the data, while the decoder architecture, in turn, shapes the features extracted from the representation. These decoders excel not only in learning high-level features in specific domains (temporal [25], [28], spatial [29], [30], [31], and spectral [32], [33]), but also in exploring hybrid multidimensional features that capture the complexity of EMG data [21], [24], [34], [35], [36], [37]. For instance, Wu et al. converted high-density EMG (HDEMG) recordings into a series of images and the electrode shift appeared as a pixel shift in images. They applied a dilated convolutional neural network to achieve robust gesture classification against electrode shift [31]. In addition, Zhong et al. developed EMG graph representations and applied a spatio-temporal graph convolutional network (STGCN) to jointly capture spatial and temporal dependencies in EMG signals [36]. Despite the substantial progress achieved through the integration of decoders and representations, the success of such systems ultimately hinges on one fundamental factor: data.

Conventionally, gesture recognition systems utilized the fully supervised learning paradigm, where deep learning decoders statistically map EMG signals to given labels. The performance of these systems depends on the availability of high-quality, diverse, and well-labeled input data, ideally

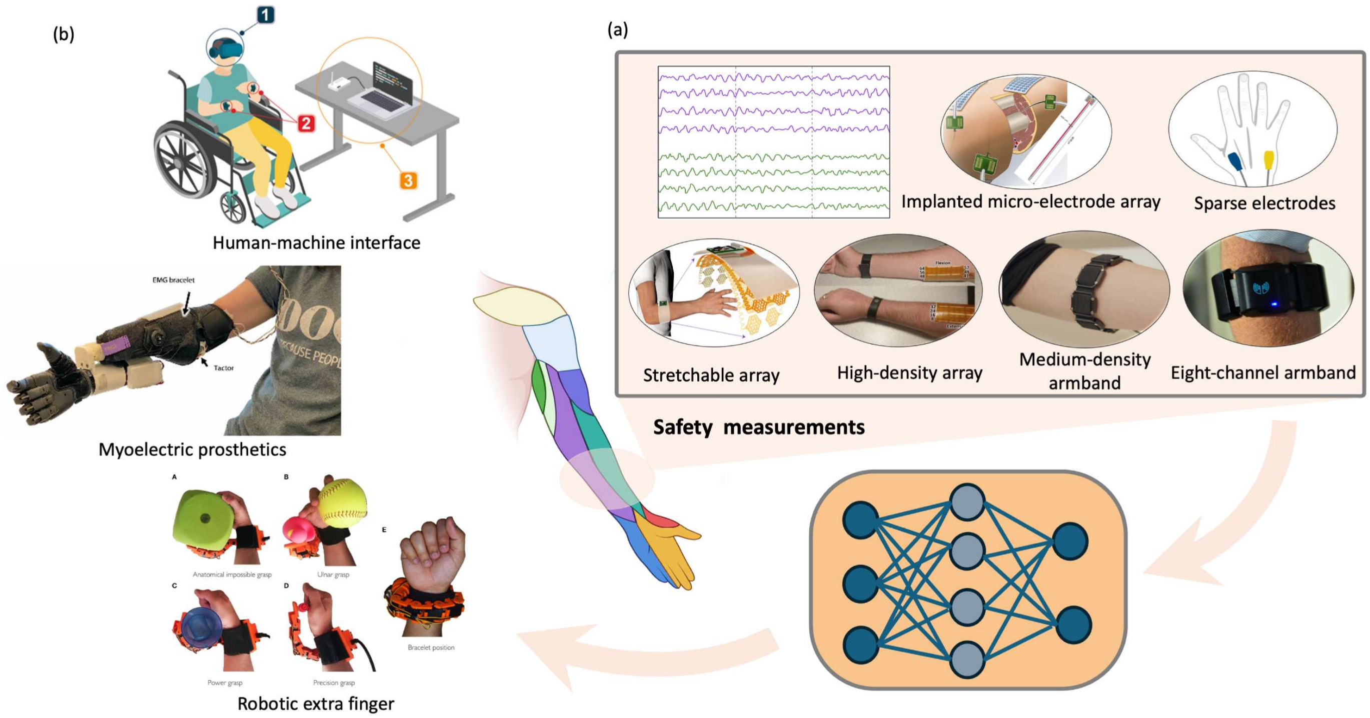

Fig. 1. Schematic representation of (a) advanced EMG sensing techniques [12], [13], [14], [15], [16], [17], [18] of EMG-based gesture recognition systems; (b) and their novel applications in human-machine interfaces [4], assistive technologies [8], and rehabilitative robotics [10].

assuming that the training and testing data are well balanced and share identical distributions [38]. In this context, Kaifosh and Reardon reported significant breakthrough with the development of a generic, non-invasive neuromotor interface capable of operating *out-of-the-box* generalization across users [5]. Their approach included recording a large sEMG dataset from 6,627 participants using a medium-density EMG wristband and supervised training of a deep neural network. This vast dataset, a scale that far exceeds typical EMG studies, allowed their models to achieve robust decoding of continuous wrist angles, discrete gestures, and fine motor tasks such as handwriting without user-specific calibration, marking a major step toward generalizable EMG control systems. This work indicates the transformative potential of large, diverse datasets in representation learning. It also highlights that such advances currently depend on labor-intensive data collection and extensive manual labeling.

Outside of such large-scale studies, however, most research and practical applications face far greater constraints. Collecting extensive, high-quality annotated EMG datasets is time-consuming and resource-intensive [5], [39], [40]. Furthermore, day-to-day usage introduces numerous sources of variability, including inter-subject validation, session-to-session differences [39], limb positions [41], and electrode shifts [42], [43], [44], would introduce distribution shifts between training and testing datasets or deteriorate the quality of EMG signals [45]. Due to these challenges, fully supervised models need to be re-trained or fine-tuned with newly collected data to maintain performance, which is unfavorable in real-world applications.

Semi-supervised learning [46] and self-supervised learning [47] paradigms have been proven transformational in domains such as computer vision [48], [49], [50], [51], [52] and natural language processing [53], [54]. These methods leverage weakly labeled or unlabeled data to enhance model generalization and adaptability, thereby reducing the reliance on extensive manual annotations. Although semi-supervised and self-supervised learning methods have demonstrated remarkable success in various domains, their application to EMG-based tasks has only recently begun to attract attention [55], [56], [57], [58], [59].

Several previous surveys have systematically reviewed machine learning techniques in EMG-based applications [27], [43], [60], [61], [62], [63], [64]. A common feature of these reviews is their model-centric focus, providing an overview of frequently used machine learning and deep learning models, often categorized by network architectures. In addition, some surveys have explored a wide range of technologies to improve model robustness and generalization of the model in EMG-based applications [65], [66]. In contrast, this survey takes a data-centric perspective, focusing on the multidimensional nature of EMG signals and their role in deep learning-based gesture recognition. Specifically, we study how different EMG representations interact with deep learning architectures during the feature learning process. We highlight the interdependence between data representation and model performance. Second, we discuss some of the data-related challenges associated with EMG data in real-world scenarios and offer insights into semi-supervised and self-supervised learning frameworks. Furthermore, we highlight the computational efficiency of

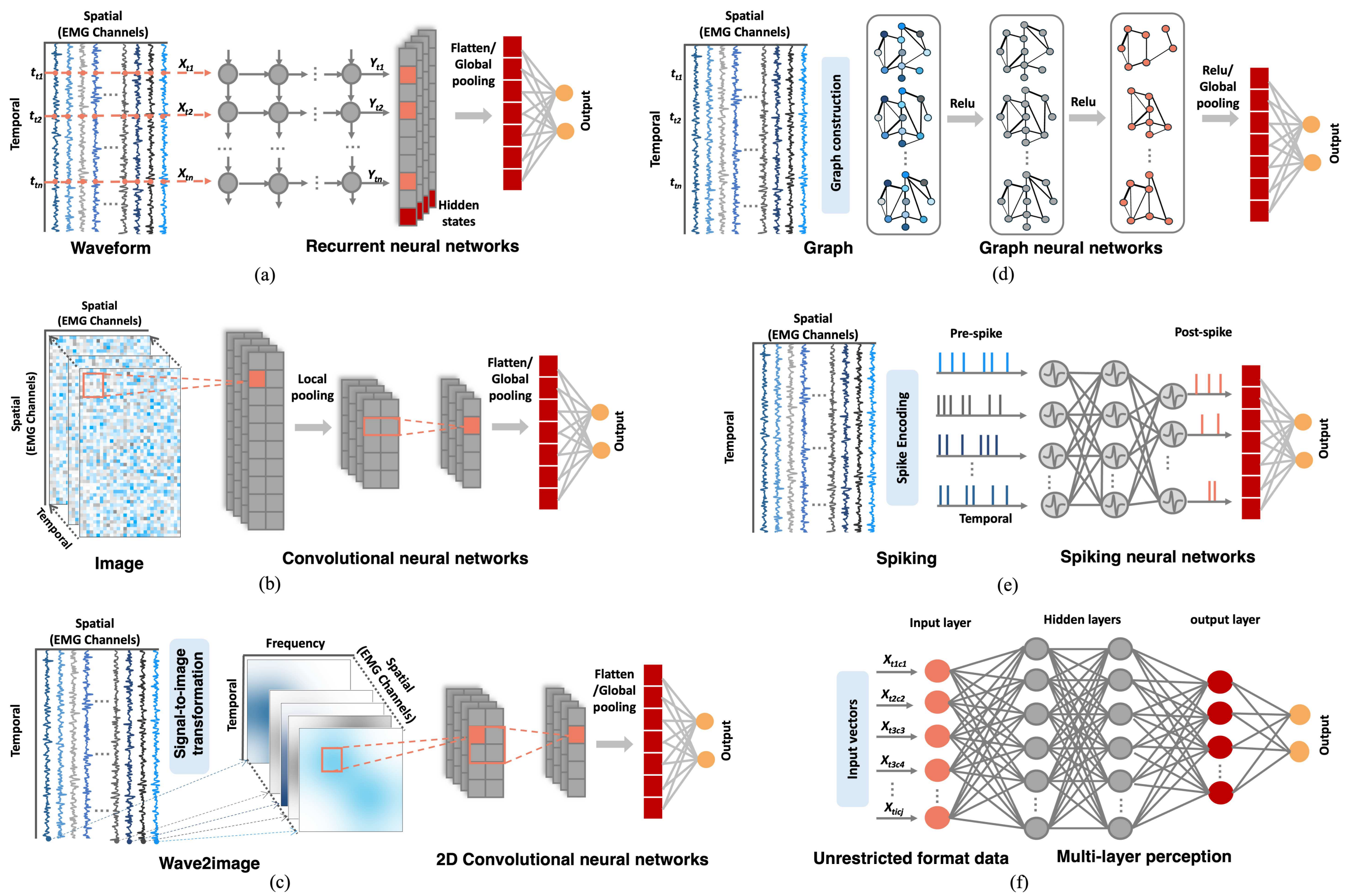

Fig. 2. Overview of six EMG signal representations and their corresponding deep learning models, following a common EMG signal filtering stage. (a) Waveform representation retains the raw temporal structure of EMG signals and is typically processed using temporal models such as temporal convolutional networks or recurrent neural networks. (b) Image representation captures spatial information across channels, making it suitable for one-dimensional, two-dimensional, or three-dimensional convolutional neural network(CNN) architectures. (c) Wave2image representation applies time-frequency transformations (e.g. short-time Fourier transform, continuous wavelet transform) to generate spectrogram-like images, which are then analyzed using CNNs. (d) Graph representation constructs graphs to model physical or functional relationships between EMG channels and employs graph neural networks for feature learning. (e) Spiking representation encodes EMG signals into binary spike trains via spiking encoding, enabling processing by spiking neural networks. (f) Unrestricted format refers to representations that disregard spatial or temporal structure, often processed by flexible architectures such as fully connected networks, multilayer perceptrons, or transformers.

EMG representation learning and examine emerging deep learning paradigms beyond traditional supervised approaches to better meet practical real-world demands.

II. Deep Feature Learning From Electromyography

Deep feature learning from EMG signals refers to the capacity of deep neural networks to automatically extract high-level and task-relevant features from EMG data, without relying on hand-crafted descriptors such as mean absolute value (MAV), root mean square (RMS), waveform length (WL), mean frequency (MNF), or median frequency (MDF). Raw EMG recordings, after standard preprocessing to remove noise, serve as the foundation for this process. They can be used directly in end-to-end learning, and many advances in gesture recognition further transform raw EMG into structured representations that emphasize its multidimensional nature across temporal, spatial, spectral, or spiking domains. These representations play a central role in guiding deep learning model design, ultimately influencing the effectiveness of gesture decoding.

In this section, we explore six major EMG representations and their corresponding deep learning architectures, as illustrated in Fig. 2. Waveform and image representations are direct structural views of the raw signal. Wave2image representations are obtained through time–frequency analysis. Graph representations leverage inter-channel relationships in multi-channel EMG. Spiking representations convert continuous EMG into discrete binary signals. Finally, unrestricted formats relax assumptions about temporal or spatial order, enabling greater flexibility for novel architectures.

A. Waveform Representation

Raw EMG signals are inherently time dependent, with each channel capturing the temporal dynamics of muscle activity as a "voltage vs. time" waveform. Waveform representations therefore preserve the signal in its most direct form, and models built on this representation typically rely on sequence-learning architectures to extract meaningful temporal patterns for gesture recognition.

Recurrent neural networks (RNNs) were applied to capture the temporal dependencies in the EMG waveforms by sequentially processing each time step [67], [68]. As illustrated in Fig. 2(a), at each time step, the current input Xt is processed along with the hidden layer $ht-1$ from the previous time step to update the current hidden state. This structure allows RNNs to retain memory over sequential data, making them suitable for learning temporal dependencies in EMG signals. However, standard RNNs struggle with long-term dependencies due to the vanishing or exploding gradient problem [69]. Long short-term memory (LSTM) networks incorporate memory cells regulated by input, output, and forget gates, allowing selective retention of information over long sequences [70]. LSTMs have shown improvements in learning long-term temporal dependencies automatically in EMG signals [71], [72], [73]. Gated recurrent units (GRUs) further optimize this architecture by merging the forget and input gates into a single update gate, reducing computational complexity while maintaining effective temporal modeling [74], [75]. Khodabandelou et al. [75] proposed an attention-based GRU framework for gesture recognition, where the attention layers capture patterns from the weights of the short term, and the GRU layers learn the interdependence of long-term temporal sequences in EMG signals.

Beyond recurrent architectures, temporal convolutional networks (TCNs) have emerged as a powerful alternative to model temporal dependencies in EMG signals [28], [76], [77], [78], [79], [80]. TCNs utilize dilated causal convolutions, where the receptive field expands exponentially with network depth, allowing long-range dependencies to be captured efficiently [81]. Unlike RNN-based models, TCNs process entire time-series segments in parallel rather than sequentially, improving training efficiency and scalability [82]. Zanghieri et al. [28] proposed an embedded gesture recognition system based on TCNs, demonstrating superior real-time decoding accuracy along with a tenfold improvement in computational efficiency compared to state-of-the-art (SOTA) EMG processing systems. A detailed summary of waveform-based approaches is provided in Table. I

B. Image Representation

Image representation builds directly on raw EMG signals by arranging multi-channel recordings into grid-like structures in Euclidean space. Each pixel corresponds to an EMG electrode location, thereby encoding spatial information across sensors. Convolutional neural networks (CNNs) extract hidden features by progressively applying spatial convolutions, integrating information from each pixel and its neighbors.

CNNs are the most widely used architecture for spatial convolution, classified into 1D, 2D, and 3D convolutions. 1D convolution [29] is suitable for low-density EMG recorded from linearly arranged electrodes. 2D convolution [83] is preferred for HDEMG, where electrode signals are mapped to a 2D layout, allowing shallow layers to capture local features and deeper layers to learn global patterns. 3D convolution [84] extends this approach by adding a third dimension to represent multiple features per electrode.

Advanced CNN variants further enhance spatial feature extraction. Multi-stream CNNs [85] divide EMG images into patches based on electrode placement, learning features independently before combining them into a unified map. Multi-scale kernel CNNs [86] extract features at multiple scales for improved accuracy. Shapley values [87] have recently been applied to CNNs to improve interpretability and accuracy driven by feedback. Furthermore, attention mechanisms [34] and vision transformers [88] have been integrated into CNNs to construct more robust spatial representations. Further related methods are summarized in Table I.

C. Wave2Image Representation

Both the spatial activation patterns of muscles and the temporal waveforms of the EMG signals carry valuable information. Spatial images typically exhibit smooth transitions and lower frequencies, whereas temporal waveforms exhibit high-frequency fluctuations over short periods. Spatial convolutions use small 2D or 3D kernels to emphasize texture, while temporal convolutions rely on larger 1D kernels to capture long-term variations [83]. Additionally, transforming waveforms into the frequency domain creates smoother spectral transitions, making them more image-like.

Building on these insights, recent studies have applied image-based models to EMG waveforms by converting them into spectrograms or scalograms via time-frequency analysis. Common approaches include the Fourier transform (FT) [94] and the short-time Fourier transform (STFT) [95], [96], [97], followed by 2D convolutional or vision transformer models [32]. Other wave-to-image methods include the continuous wavelet transform (CWT) [98], the scale-average wavelet transform (SAWT) [99], the Hilbert-Huang transform (HHT) [100], the Mel spectrogram [101], and wavelet scattering transform (WST) [102]. Additional methods are listed in Table II.

D. Graph Representation

Movement execution requires coordinated activation of multiple muscle groups to generate the necessary forces. Electrophysiological studies suggest that low-dimensional control of motor neuron modules, or muscle synergies, underlies this coordination [109], [110], [111], [112], [113]. Graph theory offers an alternative perspective, modeling EMG signals as muscle graphs, where nodes represent electrodes, and edges represent anatomical or functional connections [114], [115], [116]. This graph representation effectively captures spatial and functional relationships between EMG signals. Inspired by the success of CNNs in grid-like data, graph neural networks (GNNs) have been developed to process non-Euclidean data, preserving both network topology and node information [117], [118], [119], [120].

In muscle graph analysis, edge connections shape how information propagates across electrodes, and well-designed edge strategies improve both spatial relationship modeling and the effectiveness of gesture recognition. Redundant edges

TABLE I

EMG REPRESENTATIONS AND DEEP FEATURE LEARNING FOR GESTURE RECOGNITION: WAVEFORM AND IMAGE REPRESENTATIONS

EMG Representation	Ref.	Dataset	Subjects	Ch.	Network	Mov.	Accuracy (100%)		Real-time Plat./lat.(ms)
							Intra-sub	Inter-sub	
Waveform	[68]	CapgMyo DB-a	18 healthy	128	2SRNN	8	97.1	–	–
		CapgMyo DB-b	18 healthy	128		8	97.1	89.9	–
		CapgMyo DB-c	10 healthy	128		12	96.8	85.4	–
		Ninapro DB1	27 healthy	10		12	84.7	65.2	–
		Ninapro DB1	27 healthy	10		8	90.7	–	–
	[71]	Private	9 amputee	8	LSTM	6	91.40	–	–
	[72]	NinaPro DB5	10 healthy	16	LSTM-MSA	17	91.36	–	–
		Private 1	5 healthy	8		2	88.17	–	–
		Private 2	6 healthy	8		10	92.47	–	–
	[75]	Private	9 healthy	24	Atten-GRU	12	99.6	–	–
	[89]	Private	8 healthy	4	LSTM-RNN	5	99.00	–	–
							87.29		PC / –
	[78]	NinaPro DB1	27 healthy	10	TCN	52	89.76	–	–
	[79]	NinaPro DB1	27 healthy	10	TCN	52	74.42	–	–
							75.27		GPU / 118.54
	[28]	NinaPro DB6	10 healthy	14	TCN	8	54.5 / 49.6[1]	–	GAP8 / 12.8
		private	3 healthy	8		9	97.1 / 93.7[1]	–	
Image	[90]	NinaPro DB1	27 healthy	10	2D CNN	52	66.59	–	–
		NinaPro DB2	40 healthy	12		49	60.27	–	–
		NinaPro DB3	11 amputee	12		49	38.09	–	–
	[29]	CapgMyo DB-a	18 healthy	128	2D ConvNet	8	89.3[3]	–	–
		CLS-HDEMG	5 healthy	192		27	55.8[3]	–	–
		NinaPro DB1	27 healthy	12	1D ConvNet	52	65.1[3]	–	–
	[34]	NinaPro DB1	27 healthy	10	Atten-CNN-RNN	52	86.8-97.3	–	–
		NinaPro DB2	40 healthy	12		49	82.2-97.6	–	–
		BioPatRec26MOV	17 healthy	8		26	90.0-97.7	–	–
		CapgMyo DB-a	18 healthy	128		8	>99.5	–	–
		CLS-HDEMG	5 healthy	192		27	94.5-96.1	–	–
	[91]	NinaPro DB5	10 healthy	8	2D ConvNet	18	68.98	–	–
	[85]	NinaPro DB1	27 healthy	12	Multi-stream CNN	52	85.0	–	–
		CLS-HDEMG	5 healthy	192		27	90.3[3]	–	–
		CapgMyo DB-a	18 healthy	128		8	89.5[3]	–	–
	[84]	CapgMyo DB-a	18 healthy	128	3D CNN	8	92.3	–	–
		CLS-HDEMG	5 healthy	192		27	67.0	–	–
	[30]	Private	1 healthy	32	2D CNN	8	98.15[3]	–	NVidia Jetson Nano / 5
	[35]	Private	18 healthy	128	CNN-LSTM-TL	30	–	93.73	–
		Private (test set)	10 healthy	128		10 new	–	97.34	–
		CapgMyo DB-a (test set)	18 healthy	128		8	–	94.57	–
	[86]	NinaPro DB6	10 healthy	16	TL-MKCNN	7	97.22	–	–
							–	74.48	–
							90.30[2]	–	–
	[83]	Hyser + Private Prosth. Con.	41 healthy	1-16	2D CNN	10	–	63.39-86.33	–
		Hyser + Private Armband. Con.		32			–	65.53-75.40	–
	[92]	Private	10 healthy	8	1D CNN	21	82.93	–	–
	[93]	NinaPro DB2	40 healthy	12	CNN-VIT	49	83.05	–	–
		DB-Myo	10 healthy	8		12	90.40	–	–
	[87]	NinaPro DB1	27 healthy	10	SV-SGR	23	79.56	–	–
		NinaPro DB2	40 healthy	12		23	79.48	–	–
		NinaPro DB3	11 amputee	12		23	57.69	–	–

Ch.: number of EMG channels. Mov.: number of movements. Plat.: digital platform for real-time systems. Lat.: latency for inference. [1]Intra-session / inter-session. [2]Cross-day testing. [3]Majority voting.

can degrade performance by introducing noise and overfitting. Edge construction methods include fixed distance thresholds [105], [121], correlation-based measures [36], and learned adjacency matrices [17], [37], [106], [108]. For instance,

TABLE II

EMG REPRESENTATIONS AND DEEP FEATURE LEARNING FOR GESTURE RECOGNITION: WAVE2IMAGE AND GRAPH REPRESENTATIONS

EMG Representation	**Ref.**	**Dataset**	**Subjects**	**Ch.**	**Network**	**Mov.**	**Accuracy (100%)**		**Real-time Plat./lat.(ms)**
							Intra-sub	**Inter-sub**	
Wave2image	[95]	NinaPro DB3 NinaPro DB4	11 amputee 10 healthy	12 12	STFT + 2D CNN	17 17	60.34 88.04	– –	– –
	[96]	Private	3 healthy 1 amputee	5 5	STFT + 2D CNN	9 9	88 37	– –	– –
	[97]	Private	50 healthy	5	STFT + 2D CNN	26	78.50	62.19	–
	[32]	Hyser UC Berkeley	20 healthy 5 healthy	256 64	STFT + CViT	34 21	93.76 87.19	– –	– –
	[103]	Private	50 healthy	8	FT + AlexNet	10	94.06	–	–
	[94]	NinaPro DB2 NinaPro DB3	40 healthy 10 amputee	12 12	FFT + 2D CNN	49 10	78.71 73.31	– –	– –
	[98]	Private	10 amputee	8	CWT + 2D CNN	6	94.49	85.70	–
	[91]	Myo Dataset Private	17 healthy 8 healthy	8 8	CWT + 3D ConvNet	7 30	98.31 95.42	– –	– GPU / 21.42
	[104]	Private	2 healthy	8	CWT + CNN-Transformer	9	99.02	–	–
	[99]	Private	3 healthy	8	SAWT + 2D CNN	6	93.89	–	–
	[100]	Private	30 healthy	4	HHT + ResNet-50	7	94.41	–	–
	[101]	Private	8 healthy 4 amputee	6 6	Mel-Spec. + 2D CNN	11 11	99.41 98.00	– –	– –
	[33]	NinaPro DB5 Custom ASR	10 healthy 6 healthy	8 8	STFT + Atten	52 8	89.56 –	– 92.23	– –
	[102]	NinaPro DB5 BioPatrec	10 healthy 17 healthy	8 8	WST + Atten + CNN	53 26	≈ 90 ≈ 95	– –	– –
Graph	[105]	Lund Uni.	20 healthy	128	SAGEConv	65	90.6	–	–
	[17]	Private	3 healthy	8	CNN-GAT	18	≈ 97	–	–
	[106]	Lund Uni.	20 healthy	128	DGTG-wGIN	65	96.63	–	–
	[37]	CapgMyo DB-a CapgMyo DB-b BandMyo	18 healthy 10 healthy 6 healthy	128 128 8	STCN	8 8 15	99.8 99.4 75.8	– – –	– – –
	[36]	Lund Uni.	19 healthy	128	STGCN	65	91.07	–	–
	[107]	Private	11 healthy	128	CNN-MSTGCN TL-CMSTGCN	17	96.9 –	68.0 92.3	– –
	[108]	Private	10 healthy	8	CovGCN	9	84.69	–	–

Ch.: number of EMG channels. Mov.: number of movements. Plat.: digital platform for real-time systems. Lat.: Latency for inference.

Lee et al. [121] proposed a graph representation to capture both spatial and temporal relationships among EMG sensors. They connected physical K-nearest neighbors and established links between the corresponding sensors across consecutive time windows.

GNNs have proven to be effective in capturing spatial dependencies with EMG graph representation. GraphSAGE, which is an inductive learning method, efficiently generates embeddings for unseen data [122]. Massa et al. applied GraphSAGE to HDEMG graphs, generating low-dimensional vector representations [105]. Graph attention networks use attention mechanisms to assign importance weights to neighboring nodes [123]. Lee et al. [17] used a graph attention network and achieved approximately 97% accuracy in recognizing eight gestures across three subjects. Hybrid frameworks such as [36], [37], [106], [107] further improved decoding accuracy by integrating spatial and temporal muscle activation patterns.

Zhong et al. [36] introduced a spatio-temporal graph convolutional network (STGCN) that combines the TCN and GCN layers to effectively capture complex localized spatio-temporal correlations in the EMG nodes during gesture tasks. Similarly, Xu et al. [107] proposed a multi-view spatio-temporal graph convolutional network (MSTGCN) with spatio-temporal attention mechanism as a basic classifier, further incorporating transfer learning and CNN modules to improve feature representation ability. This approach enables reliable gesture recognition between different users. Related studies are listed in Table II.

However, most GNN-based research is focused on HDEMG, where densely packed electrodes create detailed graph structures. In contrast, sparse EMG arrays present challenges due to limited spatial resolution, which reduces the complexity and richness of the graph representation. The ongoing research aims to develop medium-density EMG systems [15], [124],

[125], and it is conceivable that GNNs could effectively leverage these intermediate resolutions for improved gesture recognition.

E. Spiking Representation

Muscles act as a peripheral gateway to access central neural activity generated at the spinal cord level. Spinal motor neurons process and transmit motor commands via action potentials, or spikes, which are often approximated as binary discrete or symbolic representation by computational models. Invasive EMG recording sensors, such as needle, fine wire, and microelectrode arrays [126] provide fine-grained insight into neural activity. However, non-invasive EMG measurements provide a global view of motor neuron activity by recording aggregated signals from the skin surface. Despite differences in resolution and invasiveness, all EMG recordings share a common origin in the spiking nature of motor neuron activity, where information is encoded at the population level through precise temporal spike patterns [127].

In the context of spiking representations for EMG-based gesture decoding, two primary steps are involved in a neural network: input encoding and output gesture decoding. The first step focuses on converting continuous EMG signals into a discrete binary format, a process commonly known as spiking encoding. Several well-established encoding mechanisms have been developed for the input of EMG. For example, delta modulation techniques translate changes in EMG intensity into spikes based on a predefined threshold, producing zero output when the threshold is not exceeded [128], [129], [130], [131]. Furthermore, EMG decomposition methods, such as convolution kernel compensation (CKC) [127], [132], [133], independent component analysis (ICA) algorithms, and blind-source separation (BSS) techniques [134], [135], have been designed to decompose composite EMG signals into their constituent motor neuron spike trains.

The second step involves decoding the encoded spiking activity to extract gesture information. Traditional machine learning methods, including support vector machines (SVM) [132], [134], linear discriminant analysis (LDA) [133], [135], principal component analysis (PCA) [136], and multilayer perceptrons (MLP) [137], have demonstrated effectiveness in processing spiking EMG data effectively. Typically, these models require the conversion of spiking activity into continuous, non-spiking features prior to further analysis. However, this transformation can lead to the loss of critical information inherent in the original spiking patterns, which can limit the performance of the model [127].

Spiking neural networks (SNNs) have been developed as an alternative. SNNs operate directly on spike-based data to encode information over time, as presented in Fig. 2(e). SNNs fully exploit the intrinsic spiking nature of motor neurons, making them particularly suited for processing EMG data in its spiking representation. SNNs incorporating leaky integrate-and-fire (LIF) neurons are simple and efficient for modeling neural dynamics [129]. In these models, the spiking neurons operate on a weighted sum of inputs. The weighted sum contributes to the membrane potential of the neuron. Once the membrane potential reaches a threshold, the neuron will emit a spike to its subsequent connections [138]. Other SNNs for EMG spiking representations include recurrent neurons [130], kernel-based models [127], [128], and deep learning-inspired spiking neurons [131], [139]. Detailed studies are listed in Table III.

F. Representation Without a Restricted Format

Although EMG signals inherently possess rich spatio-temporal information arising from muscle contractions, some deep learning models do not strictly enforce these spatio-temporal constraints. In such cases, EMG signals are treated as unstructured data, where the spatial and temporal ordering of signal samples is considered non-essential for feature extraction. This approach enables models to prioritize global feature patterns without explicitly exploiting the sequential or spatial dependencies embedded in the data. Deep learning models such as fully connected neural networks (FCNNs) [140], MLPs [141], [142], and deep forest (DF) [143], [144] typically process sEMG signals in this unrestricted format.

Furthermore, attention-based mechanisms offer an alternative approach, replacing the need for recurrent or convolutional layers by learning global relationships between input elements through attention weights [145], [146]. Similarly, the MLP-Mixer architecture operates on input data by relying on feature mixing through MLPs. In this architecture, the input is divided into patches, which are linearly projected into tokens for further processing as non-sequential data [147].

Recent studies have introduced hybrid models that blur the line between structured and unrestricted format approaches. For instance, Shen et al. [148] proposed framework combining a CNN and an MLP-Mixer for gesture recognition. CNN first processes the EMG signals to extract features in the temporal and frequency domains. These features are then passed through the MLP-Mixer, which mixes non-ordered patches of information using token-mixing and channel-mixing MLPs to produce feature representations for gesture classification.

Similarly, attention-based vision transformer models process EMG signals by dividing them into patches with patch embedding, then adding positioning embedding to preserve the positional arrangements of the patches. Hybrid vision transformers, which integrate attention mechanisms with recurrent and convolutional layers, have also been explored to leverage the strength of structured and unrestricted data representations [149], [150]. These hybrid vision transformers combine the global vision transformer modeling capabilities of attention mechanisms with the local feature extraction strength of CNNs and the sequential modeling of RNNs, providing a flexible framework for EMG-based gesture recognition.

III. Deep Learning: Evolving Beyond Supervised Methods to Semi- and Self-Supervised Approaches

The majority of gesture recognition frameworks perform effectively by utilizing supervised learning, where deep neural networks are trained with extensive labeled EMG datasets [5]. These labels represent ground-truth gestures recorded

TABLE III
EMG Representations and Deep Feature Learning for Gesture Recognition: Spiking and Unrestricted Format Representation

EMG Representation	Ref.	Dataset	Subjects	Ch.	Network	Mov.	Accuracy (100%) Intra-sub	Accuracy (100%) Inter-sub	Real-time Plat./lat.(ms)
Spiking	[137]	Private	10 healthy	16	Threshold + MLP	6	93.3	–	CPU / <40
	[132]	Private	11 healthy	192	CKC + SVM	11	96.07	–	–
	[134]	Private	3 amputee	64	BSS + SVM	9-11	>97	–	–
	[133]	Private	11 healthy	192	mwCKC + LDA	12	94.6	–	CPU / <5
	[136]	Roshamboo	10 healthy	8	Delta + PCA	3	74	–	DYNAP / –
	[135]	Private	1 tetraplegia	150	BSS + LDA	6	78.5	–	–
	[129]	NinaPro DB5	10 healthy	16 8	Delta + LIF	12	83.17 77.23	– –	FPGA / 3.94
	[128]	Private	6 healthy	100	Delta+ Spiking CNN	6	98.78	–	–
	[127]	Private	5 healthy	384	CKC + Conv LIF	10	95	54	–
	[130]	Roshamboo NinaPro DB2	10 healthy 40 healthy	8 12	Delta + Spiking RNN	3 50	85.28 55.92	– –	DYNAP / –
	[131]	NinaPro DB5	10 healthy	8	Delta + Spike FCN	12	74	–	Kapoho Bay / 5.7
	[139]	Private	11 healthy	3	Threshold+ FC LIF	5	91.21	–	–
Unrestricted format data	[140]	Myo Dataset	17 healthy	8	Fully Connected NN	7	99.78	–	–
	[141]	Private	60 healthy	8	4-layer MLP	5	85.08	–	CPU / 3
	[142]	Private	1 healthy	8	4-layer MLP	5	99.26	–	–
	[151]	private	7 healthy	8	Stacked-sparse AE	7	98.12 / 94.51[1]	–	–
	[144]	Hyser	20 healthy	256	Deep Forest	10	84.27	–	–
	[149]	Hyser	20 healthy	256	ViT-MDHGR	11	71-92[2]	–	–
	[150]	NinaPro DB2 NinaPro DB2-B NinaPro DB2-C NinaPro DB2-D	40 healthy	12	TranHGR	49 17 23 9	86.00 88.72 81.27 93.74	– – – –	– – – –

Ch.: number of EMG channels. Mov.: number of movements. Plat.: digital platform for real-time systems. Lat.: latency for inference. [1]Intra-session / inter-session.

during data acquisition. Success hinges on optimizing network parameters to map input-output pairs, ensuring accurate predictions on unseen data. However, performance depends on large high-quality datasets that are diverse, balanced, and share identical distributions between training and test sets. In real-world applications, acquiring such labeled EMG data is costly and challenging due to annotation burdens, scenario variability, and confounding factors inherent to EMG signals. Below, we outline three critical challenges affecting model robustness and reliability: data scarcity, data quality, and data variability.

- ***Data Scarcity.*** Small datasets often lead to poor signal representation and overfitting. In such cases, models memorize patterns instead of learning generalizable features, which reduces performance in unseen data. These challenges intensify in high-dimensional spaces due to the curse of dimensionality, where sparse data make pattern learning difficult [152]. In myoelectric control, gathering an extensive and varied set of EMG data from a broad range of participants and gesture types is challenging. In addition, annotating large EMG time series for movement segmentation is labor intensive. Laboratory settings use precise equipment for gold-standard annotations, such as video cue timestamps aligned with muscle activity. However, real-world scenarios lack such an apparatus, making accurate annotation impractical. Furthermore, frequent gestures dominate daily routines, while rarer actions create class imbalances, introducing modeling biases.
- ***Data Quality.*** Signal quality directly affects a model's ability to learn effective feature encoding. The stochastic, non-stationary nature of EMG signals impacts reliability, particularly in real-world settings. Signal quality can degrade due to a range of physiological, physical, and environmental factors such as muscle fatigue, atrophy, hypertrophy, crosstalk, posture changes, and electrode contact quality and placement [153], [154], [155], [156]. Additionally, during data collection, subjects may deviate from the instructed cues, resulting in inaccurate gesture labeling during post-processing [157].
- ***Data Variability.*** Data variability introduces the distribution shifts between training and test domain, ultimately limiting generalization. Major sources of variability include inter-subject differences, variations in contraction intensity, limb position, and temporal recording differences [158], [159], [160]. In addition, EMG data are often collected and processed using different protocols across datasets and studies, including variations in sensor types,

channel counts, sampling frequencies, and processing pipelines, making it methodologically harder to generalize model across datasets. Although true independence and identical distribution remain challenging, larger and more varied datasets improve robustness and generalizability [5].

Consequently, the challenges related to EMG data, namely scarcity, quality, and variability, render fully supervised learning less feasible for practical applications, as it necessitates substantial volumes of high-quality, aligned input-output data from a consistent distribution. In comparison, wearable EMG devices are producing increasing quantities of raw data with fewer annotations. Recent research has focused on synergistic methods that leverage unlabeled datasets for enhancing both data efficiency and model generalizability. Semi-supervised and self-supervised learning are two significant methods that are especially beneficial when acquiring labeled data is either costly or scarce. Although these approaches are well established in computer vision and natural language processing [54], [161], [162], [163], [164], [165], only a few pioneering studies have explored them for EMG signals [55], [56], [57], [58], [59].

A. Semi-Supervised Learning

Semi-supervised algorithms uncover latent patterns from unlabeled examples, reducing the need for extensive labeling [46], [166], [167]. Semi-supervised model training is performed on both a small amount of labeled data and a large volume of unlabeled data. The key idea of semi-supervised learning is to define a surrogate loss that enables learning from unlabeled data. Semi-supervised learning methods can be categorized into pseudo-label; consistency regularization; generative methods; graph-based methods, and hybrid methods [167]. Please refer to [166], [167] for a comprehensive theoretical survey.

In the EMG-based gesture recognition literature, Wang et al. [55] explored pseudo-labeling techniques to improve cross-user recognition performance with a little labeled data. The basic idea of pseudo-labeling methods is to generate predicted labels for unlabeled inputs using a pre-trained model and then train the model jointly on labeled and unlabeled data. Specifically, Wang et al. proposed a Self-Training based Domain Adaptation (STDA) approach for cross-user EMG gesture recognition using pseudo-labels generated through pseudo-label iterative update (PIU). PIU uses an iterative self-training loop to generate accurate pseudo-labels on new users with category balance. Their approach outperformed alternatives with improvements of 25% over baseline, 5% over supervised domain adaptation, and 24% over unsupervised domain adaptation techniques [55].

B. Self-Supervised Learning

Self-supervised learning is a branch of unsupervised learning that reduces reliance on a vast collection of manually labeled data. In the pretraining phase, the model learn from pretext tasks, which are auxiliary tasks that automatically generate data-label pairs from all input data points, whether labeled or not [51], [168]. These tasks enable the model to produce superior-to-random initial parameters and semantically valuable representations. After pretraining, these acquired features can be applied to downstream tasks (real-world tasks), enhancing model generalizability, particularly when labeled data is limited [51], [168].

Self-supervised learning methods can be classified based on their objectives into generative, contrastive, and adversarial approaches [47]. Generative methods use reconstruction loss to perform context-related generation tasks, enabling models to learn semantic representations [53], [161]. Contrastive methods rely on a contrastive similarity metric to maximize agreement between positive paired representations while pushing apart negative pairs [54]. Adversarial methods integrate generative and contrastive learning by leveraging a generative process with a distributional divergence loss, enhancing representation learning through adversarial training [47], [169]. These approaches have been widely applied in image tasks and natural language processing, inspiring a growing body of research on self-supervised learning for EMG signals.

Lai et al. employed contrastive learning for domain adaptation leading to SOTA results with $\leq$ 50% labeled data in the target domain [56]. Recently, Raghu et al. [59] used variance-invariance-covariance regularization to train continuous dynamic data, which improved myoelectric control performance and user experience. *EMGSense*[58] tackled performance degradation in inter-user biological heterogeneity in EMG sensing and designed a framework capable of adapting to a new user in a data-efficient and label-free manner.

IV. Discussion and Perspectives

Importantly, there is a key distinction between EMG representations and traditional feature engineering. Feature engineering relies on manually designed descriptors such as MAV, RMS, or frequency indices. With traditional machine learning classifiers, while effective in gesture recognition tasks, feature engineering is limited in capturing the full complexity of EMG signals in a large scale and require considerable manual effort. By contrast, the EMG representations, including waveform, image, wave2image, graph, spiking, and unrestricted formats, are not manual crafted features but signal organizations that expose relevant structure to the deep learning networks. These representations offer complementary information on the multidimensional nature of EMG signals. Deep feature learning based on these representations enables automatic capture of hidden, task-relevant features and scales more effectively to large datasets. This highlights that deep learning models are not simply a replacement for machine learning classifiers, but represents a different paradigm in which representations and deep learning models are closely coupled, allowing the full richness of EMG signals to be exploited for gesture recognition.

Waveform representations, for example, excel in capturing sequential dependencies and dynamic patterns in raw EMG signals, making them particularly suitable for applications involving continuous motion or dynamic gestures [25]. Image representations are the most common format in literature, often associated with HDEMG systems to capture fine-grained spatial information on localized muscle activation [29]. This rich spatial encoding is particularly valuable for handling inter-subject and inter-session variability [83], improving robustness against sensor shift [31], and enabling fine gesture recognition [5], [170]. Graph representations uniquely model spatial relationships between EMG electrodes, capturing inter-channel dependencies to improve signal interpretation [17], [36], [37], [105], [106], [107]. Spiking representations with SNNs decoders enable integration with neuromorphic hardware, offering the potential for energy-efficient and biologically inspired processing in gesture recognition systems [127], [131], [136]. Hybrid representations, which combine features in the temporal, spatial, and frequency domains, have demonstrated enhanced performance by leveraging the strengths of multiple domains [34], [36], [37], [106], [107]. However, hybrid models introduce inherent trade-offs, including increased model complexity, higher model demands, and greater dependence on a large dataset.

While EMG feature learning techniques have shown promising outcomes in gesture recognition, creating truly user-friendly and practical EMG control systems outside laboratory settings persists as a challenge. Real-world implementation needs careful consideration of several factors beyond just classification accuracy. Firstly, computational efficiency is paramount for real-time functioning in wearable devices, requiring lightweight models and optimized processing. Secondly, the dependence on fully supervised learning constrains scalability due to the limited, variable, and inconsistent quality of EMG data. Therefore, future training frameworks should enable learning from diverse datasets with minimal or no manual annotations. Lastly, most assessments are still conducted offline. A robust evaluation protocol would more effectively gauge system performance in dynamic, real-world scenarios.

A. Computational Effectiveness of EMG Representation Learning

There is no universally superior EMG representation; the selection of data format transformation techniques and algorithms significantly affects computational efficiency, which is crucial for real-time myoelectric use.

For EMG representation using data transforms, wave2image representations convert EMG signals into time–frequency maps, commonly via FFT, STFT, CWT and DWT. FFT and STFT are computationally efficient and offer rapid processing; however, they lack flexibility in time–frequency resolution. CWT captures fine-grained temporal changes in non-stationary signals with better resolution adaptability but has quiet high computational demanding, which hinders real-time applications. DWT presents a good compromise with lower computational demands and acceptable resolution. [171] demonstrated that both STFT and DWT can achieve high parallel speedups and energy efficiency on a parallel Ultra-Low-Power platform. Furthermore, spiking representations encode EMG signals into spike trains, commonly using delta modulation or decomposition methods such as CKC, ICA, and BSS. Delta modulation is highly efficient, relying on simple thresholding, while decomposition methods that extract MUAPs prohibit its online applications due to the computation complexity [133], [134].

Although deep learning algorithms typically incur greater computational costs compared to traditional machine learning techniques, recent research has identified specific structural optimizations within deep learning to decrease inference latency [172], [173]. An example of this is the DeepMon approach, which implemented optimizations of convolutional layer caching, decomposition, and matrix multiplication to considerably improve speed on mobile devices [174]. As illustrated by the results in Tables I–III, deep learning models for gesture recognition can fulfill the latency criteria necessary for real-time operation on embedded platforms. In particular, SNNs on neuromorphic chips represent a promising avenue for achieving energy-efficient artificial intelligence.

Alongside advancements in deep learning frameworks, there is a pressing need to enable their deployment on energy-constrained platforms, such as edge devices. The performance figures typically reported for deep learning models (see Sections II and III) are usually obtained on computers or servers equipped with GPUs, which can handle computationally intensive workloads. However, directly implementing these models on edge devices remains challenging. Therefore, deployment feasibility must be evaluated in terms of both software optimization and hardware capabilities. Recent studies (Sections II and III) have explored on-device EMG-based gesture recognition using application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and microcontrollers.

Future developments in myoelectric control systems should adopt hardware–software co-design principles, integrating model design and platform constraints from the outset. This approach entails creating lightweight models optimized for embedded platforms, improving execution pipelines, and iteratively improving digital hardware architectures. Next-generation systems must deliver real-time responsiveness, adaptability, and sustained accuracy, all within the stringent energy budgets of mobile and wearable applications.

B. Perspectives on Training Frameworks

Section III provided an overview of semi- and self-supervised training frameworks as promising strategies to mitigate challenges related to EMG data scarcity, quality, and variability by reducing the dependency on annotated data. Building on this, we highlight four emerging/promising directions that could support the development of data-efficient learning frameworks.

1) Learning With Less Supervision: Supervised learning performs well under standardized training protocols with clearly defined constraints. However, its robustness often diminishes in real-world myoelectric system deployments, where task variability and confounding factors are prevalent. Leveraging large-scale unlabeled datasets offers a pathway to more adaptable solutions in such environments. Semi- and self-supervised learning approaches enable a paradigm shift from static, one-off training towards dynamic adaptation frameworks, thereby improving model flexibility and resilience. Nevertheless, only a limited number of studies have applied these techniques to

EMG signals [55], [56], [57], [58], [59]. Given that many of the key advances in semi- and self-supervised learning have emerged from the image processing domain, these methods may serve as valuable blueprints for developing less supervision-dependent frameworks using EMG image representations.

2) Active Learning: Active learning offers a data-efficient paradigm that reduces labeling costs by selectively querying the most informative samples from an unlabeled dataset. In myoelectric control systems, prior work has explored selective sample acquisition for system recalibration through uncertainty-based active learning [175]. Uncertainty-based strategies, which prioritize samples with the highest likelihood of misclassification, are widely adopted for their simplicity and low computational overhead [175], [176], [177]. To further improve data efficiency, future research could investigate integrated deep active learning frameworks that combine the representational power of deep neural networks with lightweight active learning strategies (e.g., uncertainty-based sampling). Such frameworks have the potential to sustain long-term performance with minimal annotation requirements and only modest increases in computational cost. Furthermore, coupling active learning with semi-supervised techniques can unlock the full potential of large-scale unlabeled datasets, offering a balance between adaptability and computational efficiency [178]. A comprehensive survey of active learning methods is provided in [179].

3) Transfer Learning: Real-world EMG-based deployments remain largely conceptual, since the inherent variability of EMG signals undermines prediction accuracy and limits model generalizability. Transfer learning and adaptive learning strategies have shown promise in mitigating this challenge by refining feature representations between different individuals [45]. Moreover, modular training protocols, unlike conventional one-stage training, have been shown to enhance model robustness and generalization [180]. Recalibration or fine-tuning confined to a reduced subset of parameters further increases the suitability of these approaches for resource-constrained platforms [91]. In addition, the one- and few-shot learning paradigms enable rapid adaptation to new domains with minimal data, improving the practicality of cross-domain transfer in real-world scenarios [181].

4) Effective Augmentations: The substantial variability in EMG signals, stemming from differences across users, recording sessions, and sensor placement, can markedly compromise model robustness. Targeted augmentation strategies offer an effective means to mitigate these effects by broadening and diversifying the training distribution. Physiologically inspired *rectifiers* such as robust skeleton models, facilitate the generation of more stable and invariant representations [182], while electrode shift simulations improve resilience to changes in sensor positioning [31]. Furthermore, deep generative models [183], including autoencoders, generative adversarial networks (GANs), and diffusion models, can learn the underlying distribution of EMG signals to synthesize realistic data, thereby enriching the diversity of datasets without the expense and effort of manual data collection.

C. Offline Protocols and Online Validation

Most studiesevaluate their proposed deep learning models for EMG-based gesture recognition using open-source datasets and offline protocols. These evaluations are typically performed in controlled laboratory settings, where performance is quantified using standard metrics such as accuracy, precision, recall, and F1 score. Although such protocols provide a convenient and reproducible benchmark, they offer only a partial view of the system performance. Previous research has shown that offline assessments often do not reflect real-world conditions, as they overlook user adaptive behaviors in response to non-stationarity of the EMG signal [184], [185], [186], [187], [188], [189], [190]. In practice, the characteristics of the EMG signal can change over time due to factors such as muscle fatigue, skin impedance changes, or electrode displacement, causing users to unconsciously adapt their muscle activation patterns [155]. Offline evaluations do not capture these dynamic interactions. Moreover, they fall short in assessing real-time responsiveness [190], [191], [192], [193], physiological variability across users [194], and adaptability to environmental changes [195], all of which are critical for robust, long-term deployment in practical settings.

Nonetheless, recent studies have highlighted the continued relevance of offline evaluation in the development of EMG-based neural interfaces [196], [197]. These works report a positive correlation between offline performance metrics and real-time task outcomes, suggesting that offline analysis *can* serve as a useful predictor of in-the-loop performance. In practice, the development pipeline for novel EMG decoding algorithms typically begins with offline design and validation, which is then followed by a closed-loop, real-time implementation. Conducting a rigorous offline analysis at this early stage can help identify optimal algorithmic configurations, increasing the likelihood of achieving high performance in subsequent real-time applications [197].

Standardized online evaluation methods, such as the Box and Blocks Test [193], [198], the Target Achievement Control Test (TACT) [199], and the ISO 9241-9 standardized Fitts' Law test [200], are widely recommended to assess the practical usability of EMG-based systems. However, these protocols were originally developed for earlier generations of myoelectric interfaces and less modern, single-degree-of-freedom prosthetic hands. As multi-articulated prosthetic hands and more advanced myoelectric control systems become increasingly common, there is a growing need to revisit these legacy evaluation methods and define new standardized tasks capable of capturing the performance demands of modern systems, both in laboratory and community settings. Designing such tasks is inherently challenging, as real-world trials introduce numerous uncontrolled variables, making it difficult to disentangle the contribution of the decoder's performance from the user's adaptation strategies. Nonetheless, establishing robust, context-appropriate evaluation frameworks remains a critical step toward translating laboratory advances into reliable everyday use.

While real-time, closed-loop evaluations remain the gold standard, they are often time-consuming and resource-intensive. Consequently, offline analysis continues to play an important role in guiding iterative optimization during the early stages of development. Future comprehensive validation strategies should integrate: (1) real-time performance assessment, (2) evaluation across diverse subject populations, (3) mechanisms for continuous adaptation, and (4) testing under a wide range of environmental conditions.

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